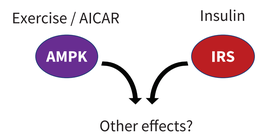
PhosR enables processing and functional analysis of phosphoproteomic data

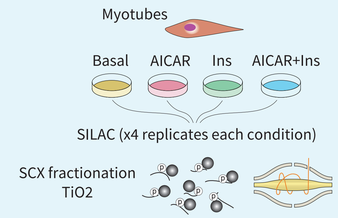
Introduction

Exercise brings many health benefits[1]. For example, muscle insulin sensitivity to glucose uptake is enhanced by exercise. Whilst this synergistic effect between exercise and insulin signalling have been demonstrated using exercise mimetics such as AICAR[2,3], the presence of other interactions between the two signalling pathways have not yet been investigated.



Experimental set up

Myotubes were treated with AMPK agonist AICAR and insulin either alone or in combination. Changes in phosphorylation against basal condition were measured by phosphoproteomics.



Processing

1. Create PhosphoExperiment class object [`PhosphoExperiment()`]

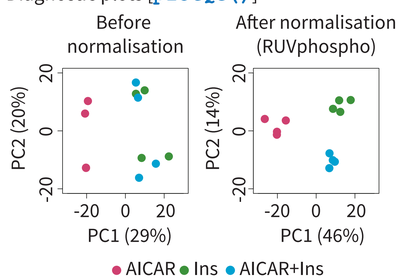
2. Filtering [`SelectGrps()`]

3. Imputation [`scImpute()`, `tImpute()`],

4. Normalisation of phosphoproteomic data for removal of unwanted variation[4]. [`RUVphospho()`]

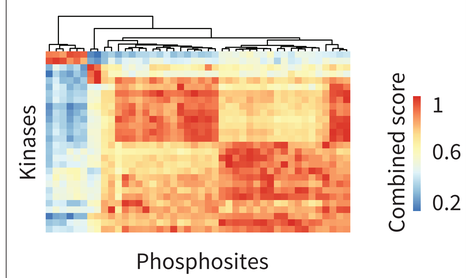
   4.1. *[Optional]* Identifying stably phosphorylated sites (SPS). [`getSPS()`]
   The SPS defined with `getSPS` can be used as a negative control in `RUVphospho`

Diagnostic plots [`plotQC()`]



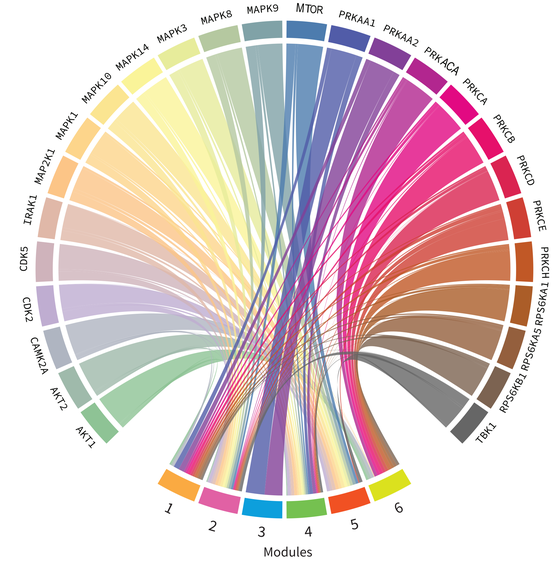
Kinase-substrate prediction

PhosR generates kinase-substrate scores based on 1) kinase recognition motif and 2) dynamic phosphorylation profiles. The scores across all kinases are integrated using AdaSampling[5] to prioritise the kinase most likely to regulate a phosphosite. [`kinaseSubstrateScore()`, `kinaseSubstratePred()`]

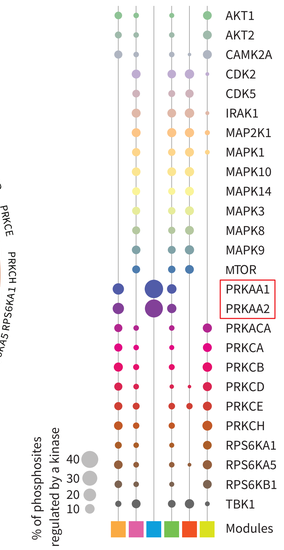


Signalling Networks (Signalomes)

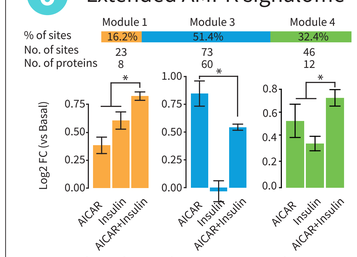
1 Signalomes [`Signalomes()`]



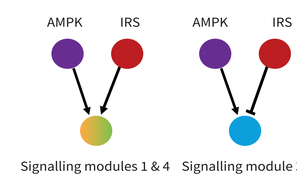
2 Signalome map [`plotSignalomeMap()`]



3 Extended AMPK signalome



4 Schematic of discordant signalling modules



Implications: Reveals a previously unappreciated interaction between AMPK and insulin signalling in coordinating insulin sensitivity.

1. We developed an approach to generate signalomes comprising discrete protein modules with phosphosites sharing similar dynamic profiles and kinase regulation. The myotube phosphoproteome stimulated by AMPK and insulin reveals six protein modules.

2. Resulting map of signalomes show that modules are regulated by different kinases and at varying proportions. Module 3 is entirely regulated by AMPK kinases, PRKAA1/2.

3. PRKAA1/2 regulate three modules, which demonstrate distinct phosphorylation profiles.

4. Whilst phosphorylation of sites in modules 1 and 4 show synergistic effects upon AICAR and insulin co-stimulation, our analysis uncovers a module wherein AMPK activity is enhanced by AICAR and attentuated by insulin.

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References
1. Ruegsegger & Booth. (2018) Health benefits of exercise. *Col Spring Harbor Perspectives in Medicine.*
2. Kjobsted et al. (2015) Prior AICAR stimulation increases insulin sensitivity in mouse skeletal muscle in an AMPK-dependent manner. *Diabetes*
3. Dokas et al. (2013) Conventional knockout of Tbc1d1 in mice impairs insulin- and AICAR-stimulated glucose uptake in skeletal muscle. *Endocrinology*
4. Gagnon-Bartsch & Speed (2012) Using control genes to correct for unwanted variation in microarray data. *Biostatistics*
5. Yang et al. (2018) AdaSampling for positive-unlabeled and label noise learning with bioinformatics applications. *IEEE Transactions on Cybernetics*

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Publications
- Kim et al. (2021) Protocol for the processing and downstream analysis of phosphoproteomic data with PhosR. *STAR Protocols*
- Kim et al. (2021) PhosR enables processing and functional analysis of phosphoproteomic data. *Cell Reports*

Web: https://pyanglab.github.io/PhosR/

PhosR

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